MULT20015 Elements of Quantum Computing

Practice Class 11

Welcome to Practice Class 11 of MULT20015 Elements of Quantum Computing.

The purpose of this lab session is to:

- understand the underpinning concepts of the Quantum Principal Component Analysis
- load a covariance matrix into a quantum circuit
- estimate the principal component with a small quantum circuit.

1 Covariance Matrix

In the lecture on quantum machine learning we saw that the first step in the quantum principal component analysis is to load the covariance matrix of our data into a quantum circuit.

In this practice class we will consider data about house sales in Melbourne. Specifically, we will look at the number of bedrooms and square footage of several houses for sale in Melbourne. The raw data for 15 houses is given by:

X = number of bedrooms = {4, 3, 4, 4, 3, 3, 3, 3, 4, 4, 4, 5, 4, 3, 4}

Y = square footage = {3028, 1365, 2726, 2538, 1318, 1693, 1412, 1632, 2875, 3564, 4412, 4444, 4278, 3064, 3857}.

For scaling purposes, we will divide the square footage by 1000.

(a) Calculate the covariance matrix for the data **X** and **Y** using the definition for covariance matrix from Lecture 21:

$$\text{Covariance (X,Y)} = \frac{\sum_{i=1}^{n}(X_i - E[X])(Y_i - E[Y])}{n-1} = \begin{pmatrix} \mathrm{Cov}(X,X) & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(Y,X) & \mathrm{Cov}(Y,Y) \end{pmatrix},$$

where E[X] and E[Y] are the mean (or expected) values of the data along the **X** and **Y** dimensions.

(b) Calculate the eigenvalues for the Covariance matrix in part (a).

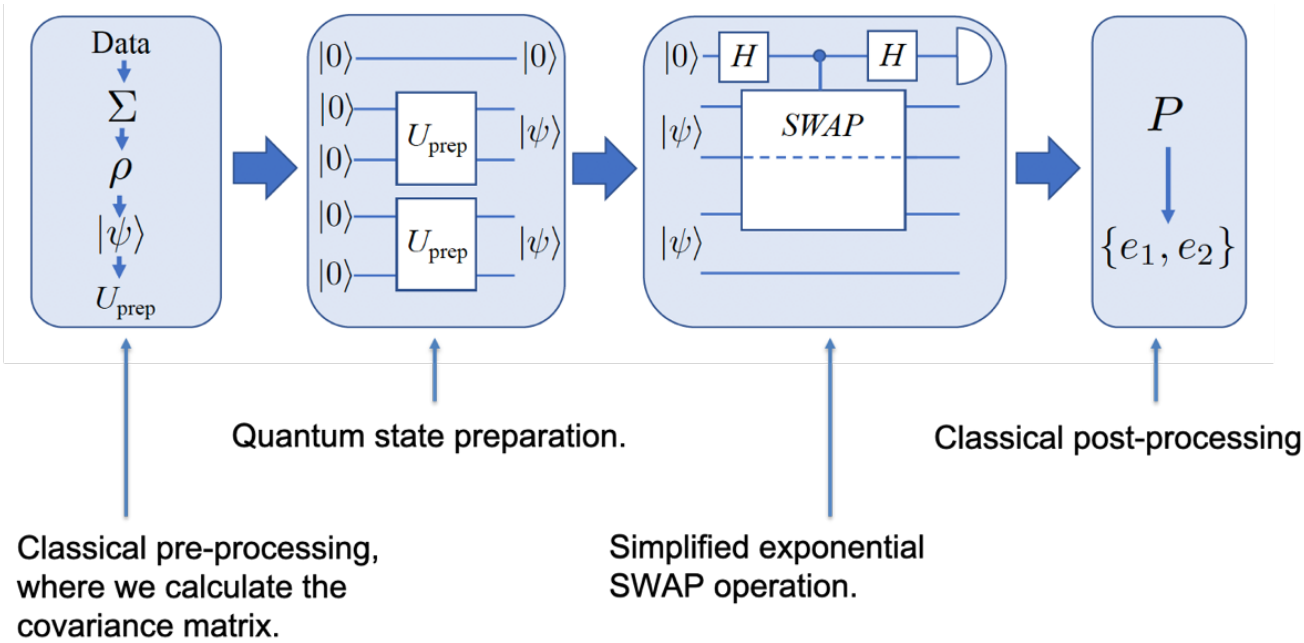
In the next question we will need to load this covariance matrix into the quantum circuit. In order for this to correspond to a physical state, we need to ensure that the trace is equal to 1.

(c) Calculate the trace of the covariance in part (a) and write down a new matrix ρ that now has a trace equal to 1

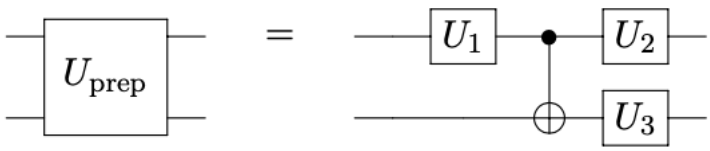
$$\rho = \frac{\text{Covariance}}{Trace[\text{Covariance}]}$$

2 Load the Covariance Matrix into a Quantum Circuit

In lecture 22 we saw that a simple way to estimate the principal eigenvalue for a set of data uses a controlled SWAP gate:



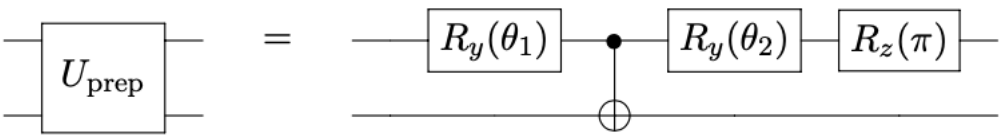
In Question 1 we calculated the matrix ρ (with trace = 1) that we need to load into the quantum circuit. In this question we will calculate the gate U_{prep} that loads the information for ρ:



The unitaries U_1, U_2 and U_3 are general single qubit unitaries and are of the form:

$U_i = R_z(\phi_i)R_y(\theta_i)R_z(\sigma_i)$, with i = 1, 2 or 3.

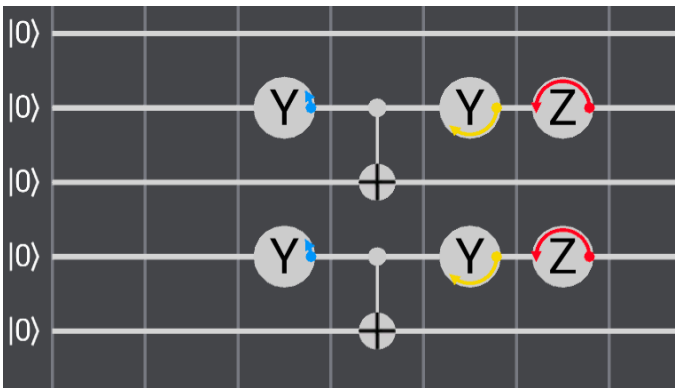
For the matrix from question 1, $\rho = \begin{pmatrix} 0.227043 & 0.341786 \\ 0.341786 & 0.772957 \end{pmatrix}$, U_{prep} can be simplified to:



(a) Solve for the angles θ_1 and θ_2 for the circuit above given the condition:

$$\rho = \begin{pmatrix} 0.227043 & 0.341786 \\ 0.341786 & 0.772957 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(1+\cos(\theta_1)\cos(\theta_2)) & -\frac{1}{2}\cos(\theta_1)\sin(\theta_2) \\ -\frac{1}{2}\cos(\theta_1)\sin(\theta_2) & \frac{1}{2}(1-\cos(\theta_1)\cos(\theta_2)) \end{pmatrix}$$

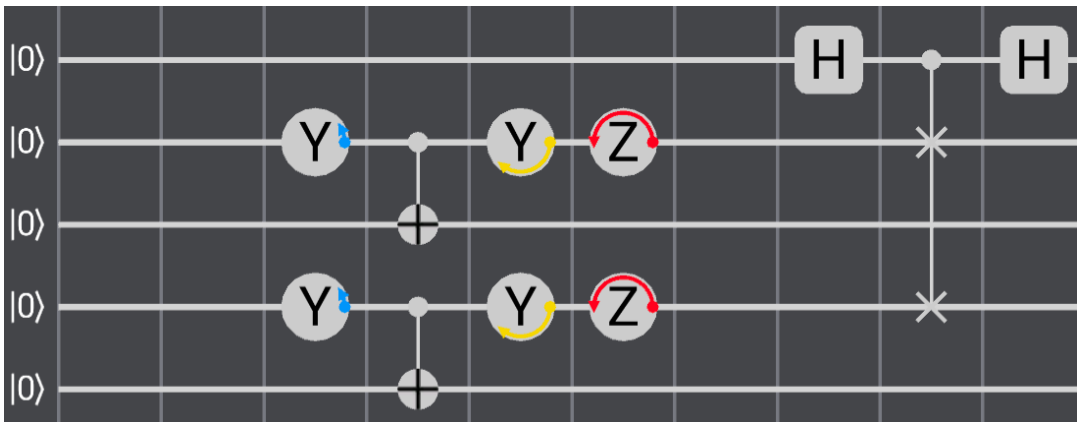
(b) Implement two copies of this U_{prep} in QUI over 5 qubits, as shown below:



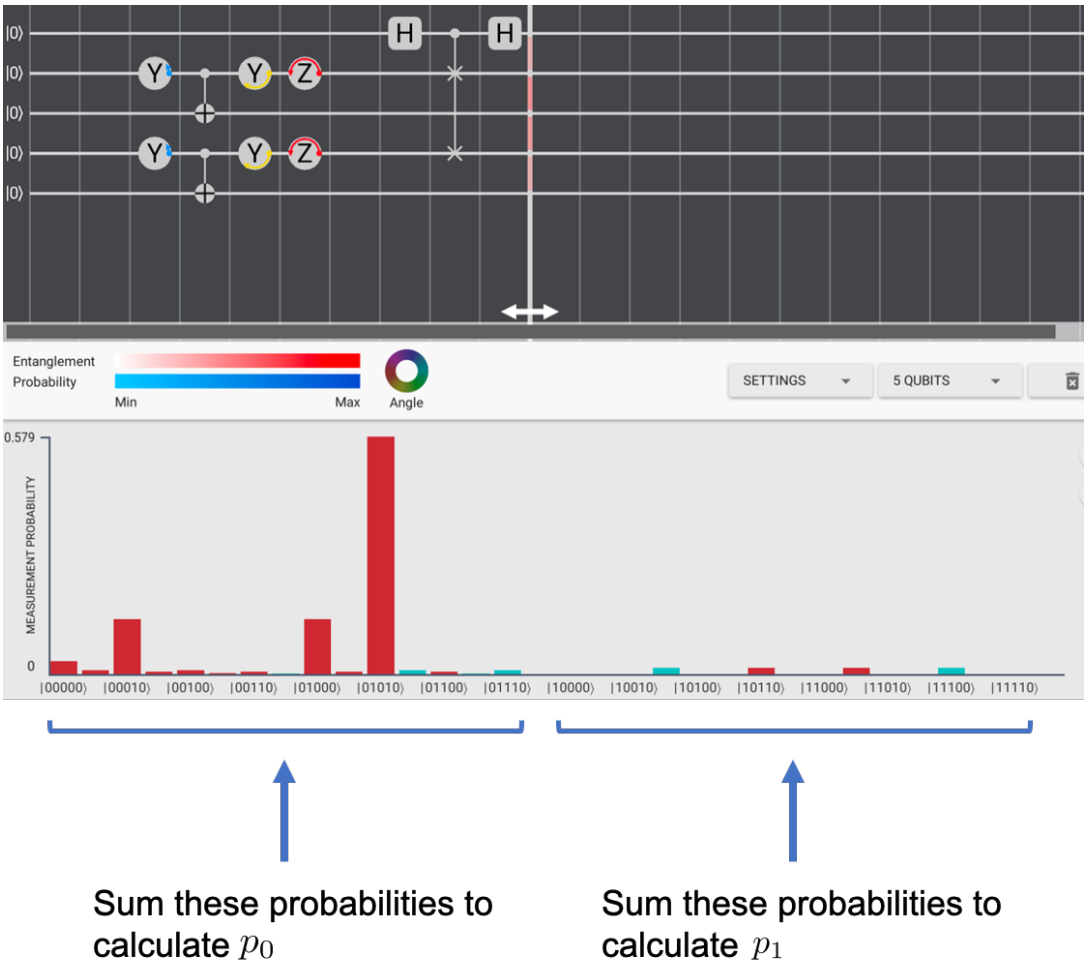
3 Estimate the Eigenvalues

In this question we will implement a controlled SWAP operation after the U_{prep} gates.

(a) In QUI, add a controlled SWAP operation, sandwiched by Hadamards, after the U_{prep} gates, as shown below:



(b) In QUI estimate the probability of detecting a 0 (p_0) and a 1 (p_1) on qubit 1. This can be done with QUI by summing up all the probabilities on the left-hand side and right-hand side of the measurement probability graph:



The estimated eigenvalues for this the Covariance matrix in Question 1 are given by:

$$\lambda_1 = Trace[Covariance]\frac{\left(1+\sqrt{1-2(1-P)}\right)}{2}$$

$$\lambda_2 = Trace[Covariance]\frac{\left(1-\sqrt{1-2(1-P)}\right)}{2},$$

where P is the difference in probability between detecting a 0 or 1 on qubit 1:

$$P = p_0 - p_1$$

(c) Estimate the two eigenvalues λ_1 and λ_2 and compare them with the eigenvalues from Question 1. Do they agree?